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Tutorial. Surface electromyogram (sEMG) amplitude estimation: Best practices

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ABSTRACT

This tutorial intends to provide insight, instructions and "best practices" for those who are novices—including clinicians, engineers and non-engineers—in extracting electromyogram (EMG) amplitude from the bipolar surface EMG (sEMG) signal of voluntary contractions. A brief discussion of sEMG amplitude extraction from high density sEMG (HDsEMG) arrays and feature extraction from electrically elicited contractions is also provided.

This tutorial attempts to present its main concepts in a straightforward manner that is accessible to novices in the field not possessing a wide range of technical background (if any) in this area. Surface EMG amplitude, also referred to as the sEMG envelope [often implemented as root mean square (RMS) sEMG or average rectified value (ARV) sEMG], quantifies the voltage *variation* of the sEMG signal and is grossly related to the overall neural excitation of the muscle and to peripheral parameters.

The tutorial briefly reviews the physiological origin of the voluntary sEMG signal and sEMG recording, including electrode configurations, sEMG signal transduction, electronic conditioning and conversion by an analog-to-digital converter. These topics have been covered in greater detail in prior tutorials in this series. In depth descriptions of state-of-the-art methods for computing sEMG amplitude are then provided, including guidance on signal pre-conditioning, absolute value vs. square-law detection, selection of appropriate sEMG amplitude smoothing filters and attenuation of measurement noise. The tutorial provides a detailed list of best practices for sEMG amplitude estimation.

1. Introduction

The surface electromyogram (sEMG) represents the spatial and temporal summation, on the skin, of action potentials propagating along the muscle fibers, where these potentials are associated with contraction of the motor units (MUs) and development of muscle force. The concept of "amplitude" of sEMG is not easy to grasp, given the random (stochastic) nature of the signal. Surface EMG amplitude refers to the voltage *variation* of the sEMG signal (see Fig. 1), mathematically defined as the signal's time-varying standard deviation. The contribution of a MU to the sEMG signal varies inversely with the distance between the MU and the detection point on the skin. As a result, a relatively thin superficial region of muscle (5–10 mm) contributes most of the sEMG signal. It is generally assumed that the activity in this portion is representative of the entire muscle. Deep muscles contribute little or no signal to the sEMG, depending on their size and depth, but may substantially contribute to the force applied to a joint, therefore disrupting the relationship between sEMG amplitude and measured torque.

The sEMG contains information about the neural drive sent to the recruited MUs, their geometrical arrangement, their conduction velocity and fatigue state, the strength of a muscle contraction, and the resulting forces exerted; but the contributions of these factors are not easy to unravel and separate. Nonetheless, the information embedded in the sEMG enables a variety of applications, including: control of prostheses (Jiang et al., 2009; Parker et al., 2006), sports medicine (Clarys and Cabri, 1993), ergonomics (Hagg et al., 2000), human machine interaction (Su et al., 2021; Xiong et al., 2021), biomechanics (De Luca, 1997), medical rehabilitation (Kumar et al., 2012; Merletti et al., 2021), clinical

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diagnostics (Kleissen et al., 1998), and biofeedback (Basmajian, 1988; De Biase et al., 2011; Giggins et al., 2013; Govil and Noohu, 2013; Ince et al., 1985). Despite promising neurorehabilitation research results and the rapid technological progress in sEMG sensor and measurement technology, sEMG has not reached the level of clinical acceptance for diagnosis or tracking outcomes that other biomedical signals (e.g., ECG and EEG) have experienced (Merletti and Muceli, 2019). Some of the barriers limiting the widespread use of sEMG in clinical applications are cultural, educational, and technical, as discussed in depth recently (Campanini et al., 2020; Merletti et al., 2021).

This tutorial is part of a series (Merletti and Cerone, 2020; Merletti and Muceli, 2019) aimed at clinical scientists and rehabilitation operators to whom this area of sEMG processing is novel, and has the objective of providing a technical background to guide the clinical use of sEMG amplitude information while warning against the most common misinterpretations of this feature. Section 2 reviews the concepts of generation of sEMG and muscle force. Section 3 recalls basic concepts discussed in previously published tutorials. Section 4 deals with signal conditioning procedures that directly affect amplitude estimates. Section 5 mentions amplitude estimation in high density sEMG (HDsEMG). Section 6 briefly discusses estimating sEMG features during electrically elicited contractions. Section 7 provides recommendations for best practices. In the future, consensus agreement on such practices might lead to standard sEMG amplitude estimation protocols and processing software; neither is currently available. Extensive references are provided herein for those who wish to further investigate this topic; consensus papers on EMG normalization and terminology are particularly important (Besomi et al., 2020; McManus et al., 2021). Mathematical expressions have been limited in favor of presentation of concepts.

2. Background

2.1. Generation of the sEMG signal and muscle force

The basic functional unit in a muscle is the MU, defined as an α-motoneuron and its associated muscle fibers (or cells). In humans, the number of muscle fibers in a MU varies between approximately 5–20 in small muscles of the eye and ear, and a few thousand in large muscles of the leg (Enoka, 1995). The fibers of a MU are dispersed throughout a

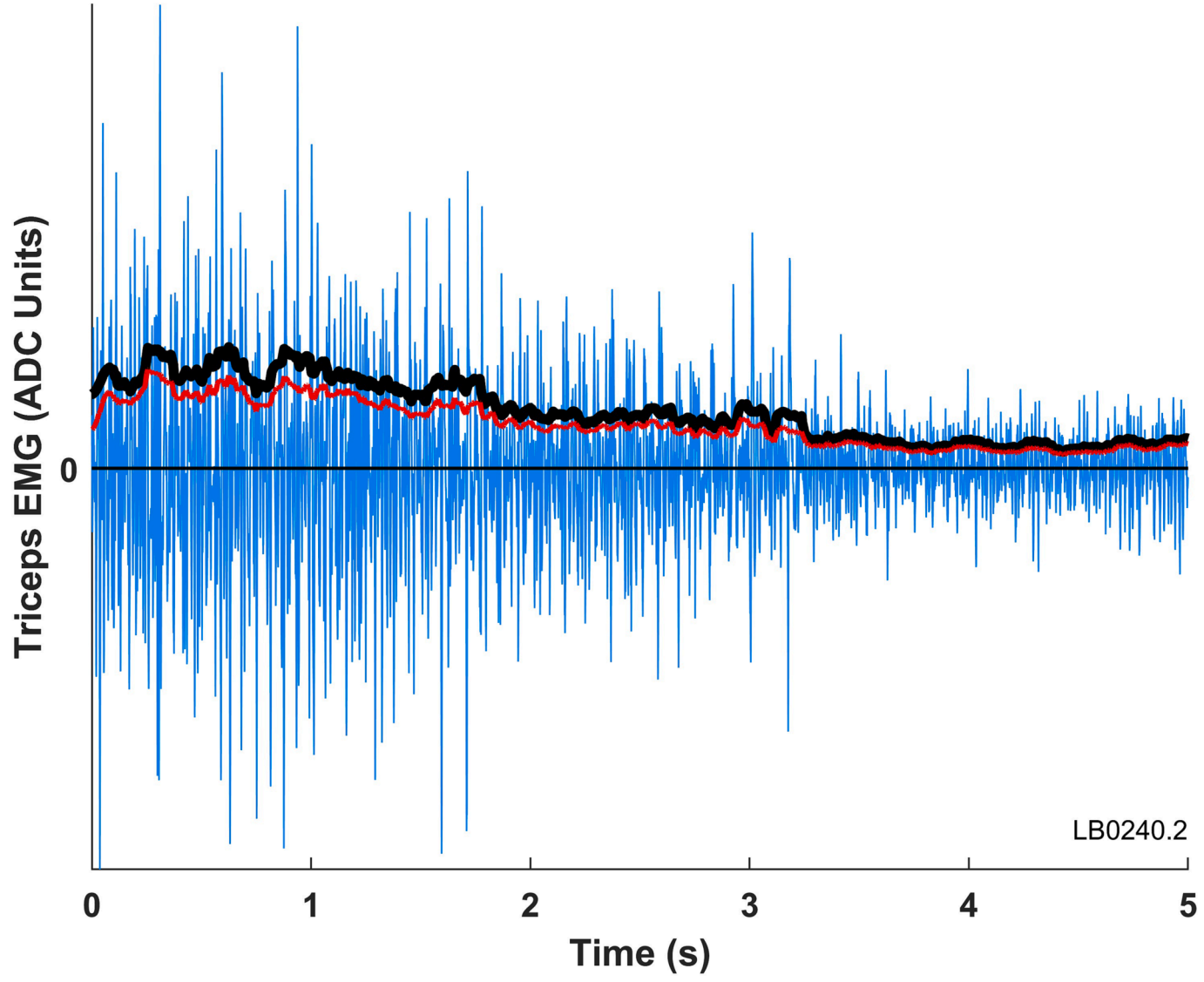


Fig. 1. Sample sEMG signal in blue, along with an RMS estimate of its amplitude (thick black line) and ARV estimate (thinner red line below the RMS plot). Surface EMG amplitude follows the general trends of the voltage *variation* of the signal, as a function of time. These sEMG amplitude estimates were produced using a 125 ms, non-causal (zero-phase) moving average window. See Section 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

portion of the muscle cross-sectional area and interspersed with the fibers of other MUs in the muscle (Monti et al., 2001). Additionally, MUs in human muscle are broadly categorized as small, slow twitch (contraction speed) type I and larger, fast-twitch type II MUs (McComas et al., 1971; Scott et al., 2001). Different fiber types contribute differently to the sEMG. The proportion of type I and type II MUs varies from muscle to muscle.

The connections between the α-motoneurons and the muscle fibers (called neuromuscular junctions) occur in a band, called the innervation zone, usually located approximately at the mid-length of the muscle (Falla et al., 2002). In a healthy muscle, each discharge of a motoneuron activates all its associated muscle fibers; the resulting single fiber action potentials travel away from the innervation zone towards the muscle tendons in both directions at a constant velocity. The propagation velocities of all fibers comprising a MU are similar. A point-like electrode on the skin surface non-invasively detects the algebraic sum of the action potentials from the individual fibers of the MU at that point. This summation is averaged in space under the area of a generic electrode and is called a motor unit action potential (MUAP), which is the basic signal that can be detected from a muscle. A single MUAP will elicit a force twitch in the muscle fibers of the MU. The duration of the force twitch (approximately 10–100 ms) is much longer than the duration of the MUAP (approximately 5–20 ms). To generate a continuous contraction, the motoneuron sends a stream of impulses over time, thereby generating a series of MUAPs called a MUAP train. If the time interval between successive MUAPs is sufficiently less than the duration of the force twitch, subsequent force twitches summate and fuse to produce a tetanic contraction which is the maximal contraction level of the MU. Active MUs do not fire strictly periodically, but at some average rate that depends upon the contraction (or force) level. During non-fatiguing, isometric, constant force contractions the firing rates of individual MUAP trains are statistically independent (Clamann, 1969; De Luca and Forrest, 1973; Englehart and Parker, 1994).

The force generated by a muscle is modulated by the neural control system, using two strategies in concert: recruitment (activating additional MUs) and rate coding (changing the rate at which active MUs fire). The exact pattern of interaction between recruitment and rate coding is muscle and task dependent (De Luca et al., 1982; Fuglevand et al., 1993). According to Henneman's "size principle," smaller, slower MUs are recruited at lower force levels and larger, faster MUs are recruited as force demand rises (Henneman, 1957; Henneman et al., 1974; Henneman and Mendell, 1981). In terms of the inter-relationship between recruitment and rate coding, active MUs modulate their firing rates in an interdependent fashion under a proposed common drive control scheme (De Luca and Contessa, 2015; De Luca and Erim, 1994).

The sEMG signal is the algebraic sum of all MUAPs generated in the detection region of the recording electrodes and averaged in space over the electrode surface. At a very low force level, there may be only a few active MUs in this region, resulting in a sparse signal recording in which the individual MUAPs may be distinguishable. As muscle force increases, more MUs are activated, and the sEMG signal becomes a complex interference pattern, such that it is not possible to visually identify the component MUAPs. As more MUs are activated at higher rates, the peak-to-peak value of the sEMG interference pattern increases with the increase in force level. Thus, a sEMG signal recorded during a moderate or strong voluntary contraction will be an interference pattern containing information on the number of active MUs, their firing rates and recruitment characteristics, and the shape of the individual MUAPs (Sanders et al., 1996); therefore, sEMG amplitude is related to muscle force (Milner-Brown and Stein, 1975) as well as to other factors.

In general—and with many caveats—sEMG amplitude varies with the force of contraction and the precise relationship may vary across muscles (Bigland-Ritchie, 1981). Amplitude modulated models of sEMG generation (Clancy and Hogan, 1994; Hogan and Mann, 1980a) and models based on composite MU firing rate and/or the number of active MUs (recruitment) (Farina and Merletti, 2001; Shwedyk et al., 1977) have been developed. Given the random nature of the sEMG signal, the static contraction sEMG amplitude estimate itself is random (or stochastic) and varies about a mean value—even during constant force isometric contractions.

As many MUAPs travel under the electrodes with random timings and different shapes, the sEMG is a random signal characterized using statistical parameters, e.g., mean and standard deviation. Since the recording instrumentation eliminates the DC component generated by imbalances of the electrode–skin interfaces, the sEMG signal has a mean value of zero. Thus, it is the signal standard deviation (often denoted by the variable σ) which gives an indication of the signal amplitude.

2.2. Emerging clinical applications of sEMG

Over the last few decades, there has been an increasing role of sEMG in clinical applications. For example, estimated sEMG amplitude from volitionally controlled skeletal muscles of spinal cord injury (SCI) patients undergoing neurorehabilitation has been employed to provide feedback to the patients and to assess treatment (Balbinot et al., 2022). Reports on the use of sEMG biofeedback in neurorehabilitation of SCI and stroke patients during walking indicated increased sEMG amplitude in some leg muscles and improvement of some gait parameters (Aiello et al., 2005; De Biase et al., 2011; Govil and Noohu, 2013). Surface EMG biofeedback was also studied in patients with upper limb nerve injuries and significant improvements in motor function were reported (Sturma et al., 2018). Recently, a Frontiers editorial project[1] on sEMG techniques for assessment and rehabilitation of motor impairment following stroke has been published (Klein et al., 2018). This special open access issue includes a total of 20 papers, covering the effects of exercise and therapy interventions on sEMG properties, mechanisms of motor impairment after stroke, and the use of novel sEMG processing techniques to study post-stroke motor control and impairment. Reported studies included an examination of the activation of the ankle muscles in post-stroke gait impairment (Banks et al., 2017) and a report on the development of a robotic arm that used sEMG to control the elbow and wrist joints, with associated neuromuscular electrical stimulation (Qian et al., 2017). Campanini et al. (2020) recorded dynamic sEMG from leg muscles of six stroke patients during walking to assess the equinus and equinovarus foot deviation deformities. They found different patterns of muscle activity across the participants, which they believe will be useful in clinical and rehabilitation treatment planning.

Clinical use of muscle synergy analysis has been suggested to improve our understanding of the underlying neural control of motor behavior patterns (Safavynia et al., 2011). Muscle synergy analysis involves the extraction of low-level motor modules (or muscle synergies) from sEMG amplitude estimates of the specific muscles involved. It is hypothesized that the nervous system employs these muscle synergies to orchestrate a variety of movement patterns through coordinated activation of the muscles (Bizzi and Cheung, 2013; d'Avella et al., 2003). Clark et al. (2010) extracted four muscle synergies during walking in healthy participants; these become merged into fewer synergies in stroke patients where the number of synergies is inversely related to the level of impairment. Muscle synergies obtained from the arm muscles of post stroke patients undergoing Wii-based therapy have also been studied (Hesam-Shariati et al., 2017). The number of synergies increased in the low functioning patients by the end of the therapy period, suggesting enhanced motor control and complexity of movement.

Since the amplitude and frequency characteristics of recorded sEMG change under different recording conditions and at different times (Halaki and Ginn, 2012), sEMG normalization is used to facilitate comparison of recorded sEMG across sessions, muscles and participants. Normalization involves expressing the signal relative to some known and repeatable value from the same muscle (Halaki and Ginn, 2012). In

[1] See https://www.frontiersin.org/research-topics/5033.

healthy participants, sEMG amplitude normalization is often achieved using the maximum voluntary isometric contraction (MVIC), which is set to a value of 100% or 1 (Burden, 2010) and the sEMG amplitude values are expressed as a fraction (or percentage) of their value at MVIC. However, producing a reliable MVIC is particularly challenging for patients, especially patients having neurological or neuromuscular disorders (Tabard-Fougere et al., 2018). The Consensus for Experimental Design in Electromyography (CEDE) has described six normalization methods (Besomi et al., 2020). Of the methods listed, normalization to a standardized submaximal contraction or to the peak or mean sEMG amplitude may be most appropriate for patient participants.

2.3. A brief review of relevant digital filtering fundamentals

Sampled sEMG is represented as a series (or list) of voltage values at equally spaced time instants, where T_s (in seconds) is the sampling period and $f_s = 1/T_s$ is the sampling frequency (in Hz). When digital filters are described, it is common notation to designate the input signal as $x[n]$ and the filtered output signal as $y[n]$, where n is the (integer) sample index. A standard "causal" filter determines the output at sample n by the addition of weighted combinations of present and past inputs and past outputs as (Proakis and Manolakis, 2007):

$$y[n] = b_0\ x[n] + \cdots + b_P\ x[n-P] + a_1\ y[n-1] + \cdots + a_R\ y[n-R], \qquad (1)$$

for constants b_i and a_i, and counting numbers P and R. The largest of P and R is the order of the filter. The initial filter outputs are corrupted, since P past input and R past output values are not all yet available, resulting in a transient. For most well-designed filters, this "startup transient" lasts less than a few hundred milliseconds. It is robust practice to discard or ignore this initial part of the output—after *all* transients have been completed. By judicious selection of the filter constants, any standard digital filter can be created, such as high-pass filters ("pass"/retain high frequencies, reject/stop low frequencies), low-pass filters ("pass"/retain low frequencies, reject/stop high frequencies,), or band-pass filters (pass an intermediate range of frequencies, reject all others). The frequency location that denotes the edge of the retained frequency band is called the cutoff frequency. In general, higher-order filters require more computation and have longer start-up transients (detrimental characteristics), but have a more ideal (sharper) shape (beneficial characteristic).

The standard causal filters of equation (1) only use present and past $x[\bullet]$ and $y[\bullet]$ values to determine each output sample. As a consequence, the output signal is always *delayed* relative to the input signal—an undesirable characteristic. There exist non-causal "zero-phase" (i.e., zero delay) filtering techniques that remove all input–output delay. The most common off-line technique to implement zero-phase filtering is the "two-pass" technique. The input signal is first filtered normally, which produces a signal delay in the filtered signal output. This output is then *time-reversed* and filtered a second time through the *same* filter. The various signal components are delayed again but, because time is running backwards, these delays are actually time advances—and they perfectly cancel the delays from the first-pass filter. This second filter output is again time-reversed, so that the final sequence is now back to sequencing forward in time. Note that a full duration filter startup transient will be found at both the head and tail of the final output signal. Hence, both head and tail portions need to be discarded/ignored. Since the designed filter has now been applied twice, the effective filter order has been doubled. Hence, if a fourth-order Butterworth filter had been designed, the effective two-pass filter is eighth order. Incidentally, this eighth-order shape is *not* Butterworth. Hence, it is best to describe what type of filter was *designed*, then state that the two-pass, zero-phase technique was used to implement the filtering process. For additional review of digital signal processing, see (Samani, 2019; Smith, 2011).

3. Recording the sEMG

Surface EMG signals can be recorded by instrumentation, where the quality of the recorded signals strongly depends on how they are acquired and processed. Therefore, researchers and clinicians must be aware of the acquisition-related issues, such as which features of hardware and signal processing software are critical for their intended purposes. Surface EMG signal acquisition requires detection using electrodes applied to the skin and signal conditioning (amplification and filtering), followed by analog to digital sampling via an A/D converter (or ADC). The peak-to-peak voltage of the sEMG signal ranges from a few μV to 10–15 mV, depending on the contraction intensity, and the signal has a limited bandwidth of approximately 10–500 Hz. As well, the recorded sEMG signal is affected by noise and several external sources of interference. Thus, sEMG signals are usually amplified by a factor of 100 to 2000 times and filtered using an analog filter, before being sampled via an ADC for input to a computer. Additional filtering and pre-processing is applied to the digital version of the signal (Merletti and Cerone, 2020).

3.1. Surface EMG detection and recording

Surface EMG signals result from the movement of ions (or charged particles) across the membranes of the muscle fibers, which produces a two-dimensional electric potential distribution on the skin that evolves over time (Merletti and Muceli, 2019). Electrodes, affixed to the skin surface overlying the muscle of interest, non-invasively transduce these time-varying potentials, which can then be measured using electronic circuitry.

Surface EMG signal quality is affected by the electrode–skin impedance, which depends on the electrode size and material, skin treatment, and electrolyte gel composition. Surface electrodes are usually made of silver (Ag), silver chloride (AgCl), gold (Au), or silver/silver-chloride (Ag/AgCl). Because Ag or AgCl electrodes are almost non-polarizable, they are often preferred over gold or gold-plated electrodes. In non-polarizable electrodes, the electrode–skin impedance is predominantly a resistance with a small capacitive component, thus the electrode–skin impedance is minimally affected by the signal frequency (Cattarello and Merletti, 2016; Merletti and Cerone, 2020). Skin preparation is performed to reduce the electrode–skin impedance as much as possible and improve the stability and the contact quality of the electrodes. This is usually done by rubbing with alcohol, or (and better) with conductive abrasive paste to remove dead skin cells and oily substances (Merletti and Cerone, 2020; Piervirgili et al., 2014).

3.1.1. Effect of electrode size and configuration

Electrode size is defined as the electrical contact area (metal or conductive gel) of an sEMG electrode. A large range of electrode sizes can be found in the literature. An electrode *samples* the signal distribution in *space*, in that the sEMG differs at each location on the skin and each electrode only *samples* the voltage under that electrode. Since each electrode transduces the average voltage under its contact surface, a *smoothing* action is applied in space and reflected in time as a low-pass filtering, therefore affecting the sEMG amplitude. This effect is similar to the observation that sea waves lift/lower a boat much less than a cork (Merletti and Muceli, 2019; Stegeman et al., 2000). Thus, a large electrode provides a smoothed version of the evolving sEMG, causes a loss of signal detail, and concurrently decreases the RMS value (Afsharipour et al., 2019; Merletti and Cerone, 2020; Merletti and Muceli, 2019). Hence, electrodes with diameter of 3–5 mm are recommended over larger electrodes (Merletti and Cerone, 2020; Merletti and Muceli, 2019).

Different electrode configurations can be used based on the application and the purpose of the sEMG recording. In the monopolar configuration, the voltage of each electrode is measured with respect to a reference electrode which is located over a region having little muscle

activity (e.g., a nearby bony region). Monopolar sEMG recordings are prone to contamination from power-line interference, activity from large and deep MUs, activity from other muscles (known as crosstalk), and other common mode noise sources, which restrict their use to only a limited set of research applications (Merletti and Hermens, 2004). To attenuate common mode interference, sEMG signals are often detected as a linear combination of the signals recorded at different electrodes. The simplest form of this filtering is the single differential (a.k.a. bipolar or SD) configuration where a pair of electrodes is aligned in the direction of the underlying muscle fibers and the difference between the two adjacent monopolar signals is taken. The common mode voltages in the two monopolar signals are greatly attenuated in the bipolar signal (Hof, 1984). The double differential (DD) configuration (which requires three electrodes) can be obtained from the difference between two adjacent single differential signals. The benefits of using differential detection are the identification of the innervation zone, and the attenuation of common mode and non-propagating signal components, which have the effect of increasing sEMG amplitude relative to these interferences (Campanini et al., 2020; Merletti and Muceli, 2019). In particular, bipolar electrodes placed over the innervation zone produce a dramatically reduced sEMG amplitude, resulting in lower signal to noise ratios (Farina et al., 2002). Such placement should be avoided, e.g., by detecting the innervation zone. More information and schematic drawings of MUAP propagation for different configurations can be found in[2] Campanini et al. (2020).

The interelectrode distance (IED)—the center-to-center distance between the conductive areas of two electrodes in a single differential configuration—strongly impacts the sEMG signal characteristics. To avoid filtering effects over the frequency range of interest of the sEMG signal (10–500 Hz), the IED should be in the range of 5–10 mm. More detail on IED effects can be found in Merletti and Muceli (2019).

3.2. Signal conditioning: Amplification, analog filtering and digitization

Signal conditioning in an analog sEMG acquisition system comprises electronics to: a) amplify the input signal to the voltage range accepted by the ADC (typically 3–10 V), b) filter out frequency components outside the bandwidth of the sEMG, and c) reject common mode interference sources (e.g., power-line interference) within the frequency band of the signal (Merletti and Cerone, 2020).

3.2.1. The front-end amplifier: Main features and parameters

The amplifier is an analog electronic circuit usually comprising a front-end differential amplifier and a series of stages to further amplify the signal from mV levels to peak-to-peak values of a few volts, and filter unwanted frequencies. Some critical features of the front-end differential amplifier that a clinical user should be aware of are briefly reviewed here and fully defined elsewhere (Merletti and Cerone, 2020; Merletti and Hermens, 2004).

Input impedance (Z_i): The input impedance of the front-end amplifier is the opposition to current flow into the amplifier—as a function of frequency—between each input and the reference (a.k.a. zero voltage). The input impedance of a sEMG amplifier should be higher than 100 MΩ at 50 or 60 Hz, but 1 GΩ is preferable in case of small dry electrodes (Merletti and Hermens, 2004; Tankisi et al., 2020). A high value of Z_i is important for rejecting common mode voltages (Merletti and Cerone, 2020).

Differential and common mode amplifications: The input signals to a differential amplifier contain two components: the common mode (V_{cm}) and the differential mode (V_{diff}) signals. The common mode signal is the average of the two input signals, while the differential mode is the difference between them. Each of these signals has their own amplification factor: A_{cm} and A_{diff}, respectively. The ratio A_{diff}/A_{cm} is called the common mode rejection ratio (CMRR), which describes the amplifier's capability to reject or attenuate common mode voltages (in particular, the power-line voltage); the acceptable minimum CMRR value is 90–110 dB at 50–60 Hz (Merletti and Cerone, 2020; Merletti and Hermens, 2004; Tankisi et al., 2020).

Front-end amplifier noise: A front-end amplifier generates electronic noise whose power is uniformly distributed across all frequencies (white noise) with additional contributions below 10–20 Hz (flicker noise) at the lower edge of the sEMG bandwidth (Merletti and Hermens, 2004). These noise sources, which are different from interference (e.g., ECG, power-line interference) or crosstalk, typically add to the signal of interest and limit the smallest sEMG signal level that can be detected (Merletti and Cerone, 2020; Merletti and Hermens, 2004). This noise is usually smaller than the noise generated by the electrode–skin interface, which is more critical.

3.2.2. Filtering

The front-end amplifier is followed by analog filter(s) before converting the detected sEMG into digital form for further processing and storage. Signal filtering refers to attenuation of specific frequencies in a signal which are outside the bandwidth of interest, or sometimes inside this bandwidth such as the case of power-line interference. The bandwidth of the sEMG signal generally ranges from 10 Hz to 500 Hz.

In a high-pass filter, frequencies below the cutoff frequency are greatly attenuated, while the frequencies beyond that pass through the filter. This type of filter is used to attenuate any DC (or 0 Hz) and low frequency (or slowly varying) components of a detected signal, often including motion or cable artifacts, which are generally below 10 Hz (Tankisi et al., 2020). High frequency noise is reduced using a low-pass filter with a cutoff frequency above which the frequencies are attenuated; the low-pass cutoff frequency for sEMG is normally in the range of 450–500 Hz. Power-line interference at 50 or 60 Hz can be attenuated using an analog band-reject or notch filter, which attenuates frequencies in a small frequency band about the notch filter frequency (Merletti and Cerone, 2020), however, digital filters are preferred for this last purpose. As not all interference is removed by the analog filtering stages, digital filtering is generally done as a first processing step post-sampling as discussed in Section 4.1.

3.3. Analog-to-digital conversion

The amplified and filtered sEMG signal is an analog signal, whose continuous value varies over time. To be stored and processed on a computer, this signal must be sampled to a digital format, which is done by means of an ADC. Sampling is generally done at periodic intervals (i.e., at a constant sampling rate). The sampling process—the transformation of the analog signal from the continuous domain to a sequence of samples—can result in a loss of information about the signal between samples, if the sampling frequency is not sufficiently high. For a sEMG signal with a recording bandwidth of 500 Hz, the sampling rate must be at least 1000 Hz based on the Shannon-Nyquist criterion (Blinowska and Zygierewicz, 2022); however, a sampling frequency $\geq$ 2000 Hz is recommended for better signal fidelity[3] (Merletti and Cerone, 2020). For more detail and information regarding ADC characteristics (i.e., resolution, quantization errors, and sampling rate effects), refer to Merletti and Cerone (2020).

4. Estimating sEMG amplitude

As noted above, interference-pattern sEMG is well modeled as a random process (Clancy and Hogan, 1999, 1995; Hogan and Mann,

[2] See also: https://www.robertomerletti.it/en/emg/material/teaching/module6.

[3] See also: https://www.robertomerletti.it/en/emg/material/teaching/module6.

1980a, 1980b; Shwedyk et al., 1977). We wish to obtain sEMG amplitude estimates that minimize errors and variability. Thus, we refer to this processing as sEMG amplitude *estimation*. In most applications, contraction effort is changing over time, thus so will the sEMG amplitude estimate.

Surface EMG amplitude estimation (from the acquired digital sEMG signal) consists of at least five stages (Fig. 2, Fig. 3): a) pre-processing, in which unwanted residual noise/interference is removed from the acquired sEMG signal; b) detection in the form of either an absolute value (a.k.a. rectification) or squaring operation; c) low-pass filtering with a cutoff frequency $\leq$ 10–20 Hz (and typically closer to 1–2 Hz, depending on the application); d) if the detector utilized a squaring operation, the square root is taken, returning the signal to its original physical units of voltage; and e) additive noise correction due to background noise (e.g., electrode–skin interface noise and electronic instrumentation noise). It is recommended to record resting sEMG before each measurement to document this background noise level. The following sub-sections describe each stage in more detail.

4.1. Pre-processing: Removal of unwanted noise/interference sources

Invariably, some amount of noise/interference is acquired along with the sEMG signal (Clancy et al., 2002; Merletti et al., 2016). This noise is mostly generated at the electrode–skin interface and may have a RMS level of 1.1–4.5% of the sEMG RMS level at MVIC (Wang et al., 2019). Some of this noise occurs over a range of frequencies within the same band as the sEMG. Other noise sources either occur mostly outside of the sEMG frequency band (e.g., motion artifact) or over very narrow frequency spans (e.g., power-line interference). Thus, further "pre-processing" or "signal conditioning" filters are applied to the digital sEMG signal to remove these undesired components (Merletti and Cerone, 2020).

4.1.1. Pre-processing: High-pass and low-pass filtering

The primary goals of high-pass filtering are to eliminate DC offsets from the acquired signal and reduce movement artifacts. These undesired sources occur at frequencies $\leq$ 10–20 Hz (McManus et al., 2021). Since there is little or no usable sEMG in this frequency range, a high-pass filter is applied with cutoff frequency set between 10 and 20 Hz (De Luca et al., 2010). The precise cutoff value is not critical (Merletti et al., 2010). Filters of order 2 or 4 are common. Filters of order as high as 6 or 8 are not unreasonable for applications in which large motion artifact is expected (e.g., jumping, hopping or vibrating a limb), but must be carefully designed and implemented to avoid excessive startup transients that can introduce "ringing" artifacts during rapid signal transients [see Figure 16.5 in (Merlo and Campanini, 2016)]. Conventional filter types (e.g., Butterworth or Chebyshev) are typically selected. Fig. 3A shows an acquired sEMG signal in which an obvious motion artifact has been artificially inserted near time 4 s. This sEMG signal was high-pass filtered by designing a second-order Butterworth filter with a cutoff frequency of 10 Hz, then filtering using the two-pass zero-phase technique. Fig. 3B shows that the motion artifact is essentially eliminated by the high-pass filter, with no other noticeable changes to the signal—as desired.

Most of the power in the acquired sEMG signal exists below a maximum frequency of 400–500 Hz (Gallina et al., 2022; Merletti, 1999), which requires a sampling rate of at least 1000 Hz. But, a faster sampling rate (e.g., $\geq$2000 Hz) is desirable (Gallina et al., 2022). Hence, including an analog low-pass filter prior to the ADC is common, typically with a cutoff frequency of 450–500 Hz. If necessary, the sampled signal can be further low-pass filtered digitally, using one of the conventional filter types of order 2–4.

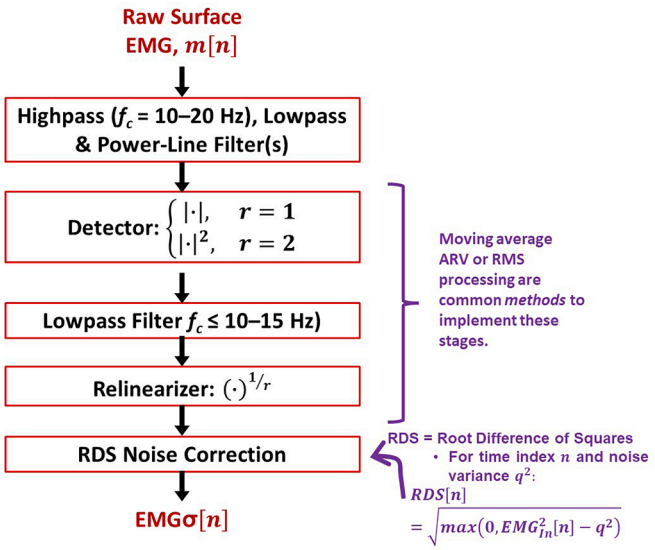


Fig. 2. Surface EMG amplitude estimation steps. 1) Raw sEMG are bandpass filtered (high-pass filter cascaded with a low-pass filter). Power-line filter(s) may be applied. 2) Detection either using an absolute value (i.e., rectifier) or squaring operation. 3) Low-pass filtered with cutoff frequency $\leq$ 10–15 Hz. Moving average filter serves as the low-pass filter in ARV or RMS processing. 4) If square-law detector was used, square root is applied. 5) Additive noise correction. Since the output sEMG amplitude estimate varies much more slowly than the raw sEMG input signal, downsampling may be applied to the output.

4.1.2. Pre-processing: Filtering of power-line interference

A notch filter can be used to attenuate power-line interference (Fig. 4). A notch filter has a gain near zero at the interference frequency, but a gain near one at all other frequencies. Such filters are extremely effective in removing interference at 50/60 Hz. Filters can easily be designed with notch widths of 1 Hz (Ranaldi et al., 2022), excising power-line interference with limited impact on other frequencies (Merletti and Cerone, 2020). Notch filters can also be applied at multiples of the power-line frequency (i.e., harmonics), if necessary. For computational reasons, notch filtering at multiple harmonic frequencies should be implemented in cascade.

Alternatively, the discrete Fourier transform (DFT) can be used to remove power-line interference (Mewett et al., 2004). The signal is converted into the frequency domain using the DFT (to obtain the frequency spectrum), a small frequency span of values centered about the interference frequency is removed and replaced by interpolation of previous and subsequent spectral values, then the inverse DFT of the resulting spectrum is computed. While both startup and tail filter transients will exist, the magnitude of these transients tends to be quite small. Further, if the signal is not affected by substantive power-line interference, then the EMG signal is barely altered. This process should only be applied to the *entire* signal and individual epochs should be defined afterwards. See Fig. 4E, F for an example. The process may be repeated for harmonics of the power-line, if present. See (Merletti and Cerone, 2020) for additional detail. Other methods to attenuate power-line interference include time-adaptive noise cancelation (Martens et al., 2006) and subtraction of a best-fit sinusoid at the interference frequency from the entire signal. Wireless transmission substantially reduces this problem.

4.1.3. Pre-processing: Other filtering

Two examples of other filtering will be given herein. First, additional interference sources, recorded at specific frequencies within the sEMG band, can originate from external devices. It is, therefore, common to review the frequency content (i.e., power spectral density) of sample recordings for the existence of consistent interference at specific localized frequencies, usually selecting rest recordings when doing so. If unwanted interference (sharp spectral peaks due to other equipment) is found at specific frequencies, notch filters or spectral interpolation can

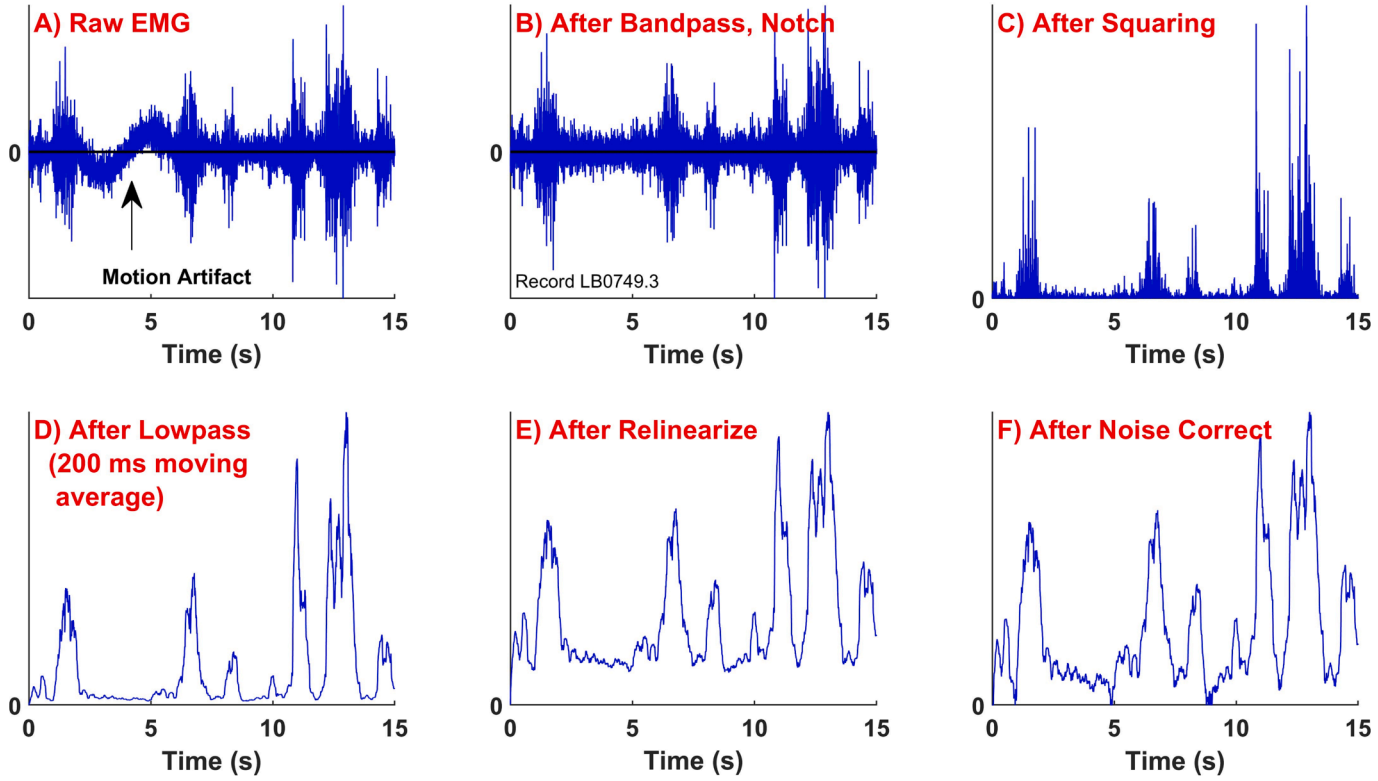


Fig. 3. Sample time series of one sEMG signal at each major stage of processing (major stages shown in Fig. 2). Data recorded over a biceps muscle during constant-pose, force-varying contractions ranging from 50% MVIC extension to 50% MVIC flexion. See text for tutorial description of each stage. Each y-axis is auto-scaled.

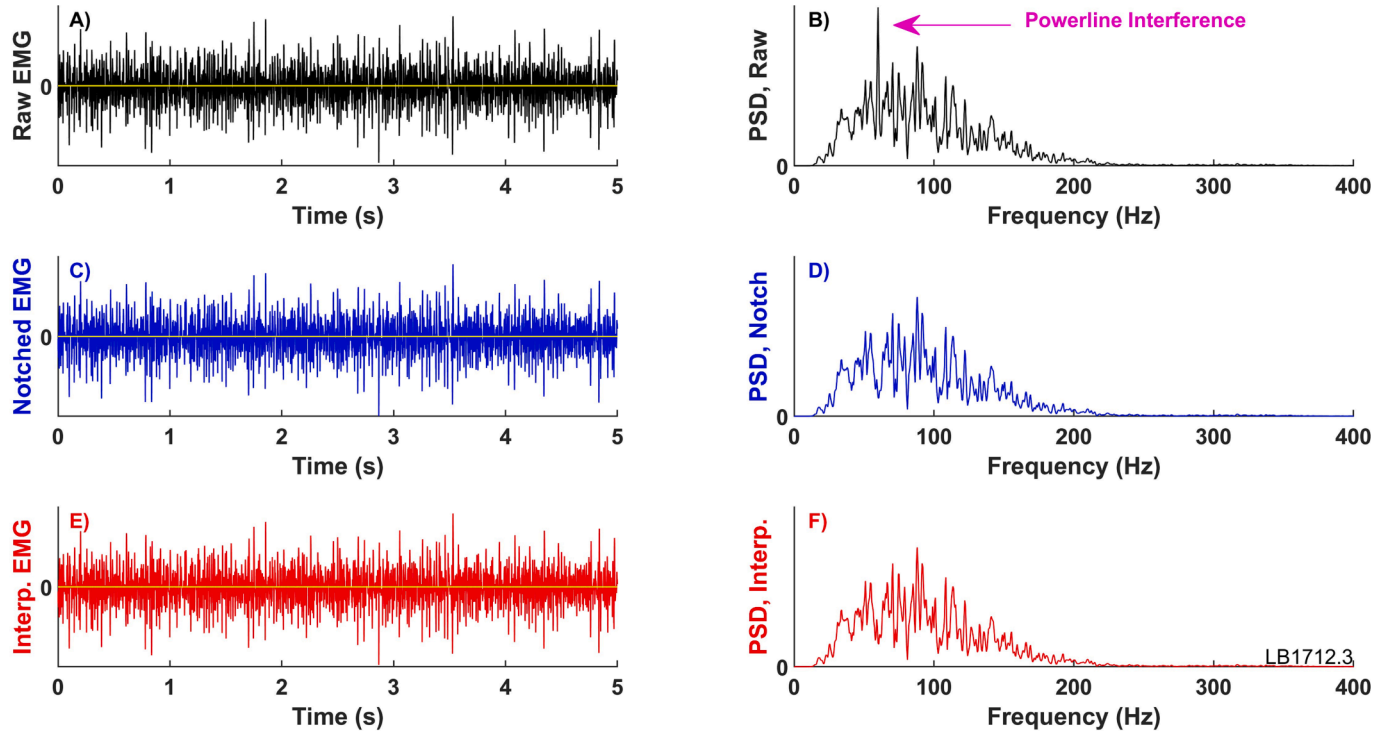


Fig. 4. Use of power-line notch filter. A) Raw sEMG during constant-posture, constant-force, 50% MVIC contraction. B) Corresponding power spectral density (PSD) estimate, showing obvious 60 Hz power-line interference. C) Surface EMG signal after notch filter with bandwidth of 1 Hz. D) PSD estimate of notch filtered sEMG; notice absence of power-line interference. E) Surface EMG signal after frequency interpolation method (interpolated over 59.5–60.5 Hz) for removing power-line interference. F) PSD estimate of frequency-interpolated sEMG; again, notice absence of interference. Axes scales are the same for all time-domain plots and, separately, for all frequency-domain plots. Note that notch and interpolation bandwidths smaller than 1–2 Hz should be applied with caution, since the power-line frequency can fluctuate more than this amount, especially for the higher harmonics of the power-line frequency.

be applied to reduce it.

A second common interference source is other physiological signals. For example, sEMG recorded from the torso can easily be corrupted by ECG (Barrios-Muriel et al., 2016). The ECG signal power (mostly below 30 Hz) overlaps the lowest portion of the sEMG frequency range. Nonetheless, some investigators have increased the pre-processing sEMG high-pass filter cutoff frequency to ~ 30 Hz and used high-order filters (4th–8th order) to diminish ECG power while limiting loss of sEMG power (Drake and Callaghan, 2006; Redfern et al., 1993). Numerous more advanced de-noising techniques have been developed (Chen et al., 2016; Lu et al., 2009; Mak et al., 2010). Unfortunately, a standard pre-processing procedure has not yet been proposed.

4.2. Detection

The sEMG signal mean value is forced to zero by high-pass filtering. The voltage *variation* of the sEMG signal is modulated by muscle effort and is the quantity of interest. This variation is quantified as the signal standard deviation, or sEMG amplitude, as a function of time. Surface EMG average rectified value (ARV) and root mean square (RMS) are *methods* for estimating sEMG amplitude. All such methods estimate EMG amplitude using a brief time interval of the raw EMG signal.

Mathematically, a "detector" is used as the next stage in estimating sEMG amplitude. Either an absolute value operation (a.k.a. rectifying or rectification) or a squaring is used. In practice, either detector is acceptable, and the performance differences between them is small (Clancy and Hogan, 1999). If $y_1[n]$ denotes the output of the pre-processing stage and $y_2[n]$ denotes the output of the detection stage, then an absolute value detector[4] is defined as:

$$y_{2,abs}[n] = |y_1[n]|, \tag{2}$$

and a square-law detector is defined as:

$$y_{2,square}[n] = y_1^2[n] = |y_1[n]|^2. \tag{3}$$

When combined with an averaging filter as the ensuing low-pass filter stage (discussed below), the square-law detector has a formal relationship with signal energy, which can be a useful attribute. When sEMG processing was more commonly completed in analog hardware, the absolute value detector was more common, since it could be implemented using diodes (a.k.a. rectifiers). After detection (Fig. 3C), the signal is now only positive-valued.

4.3. Low-pass filtering (a.k.a. smoothing)

We desire to retain the very low frequency portion of the detected signal (i.e., the signal trend), and attenuate the higher frequency fluctuations as much as possible. Thus, an additional low-pass filter is used (Fig. 3D). Either moving-average or conventional low-pass filters are used.

4.3.1. Moving average and conventional low-pass filters

Theoretically, the optimal low-pass filter has only been derived when the contraction is constant-pose, constant-force and non-fatiguing (Hogan and Mann, 1980a, 1980b). In this case, if the load sharing among the active muscles is not changing, the *true* sEMG amplitude is not changing throughout the contraction[5] and all samples of the detected sEMG should contribute equally to the sEMG amplitude estimate. Thus, a moving average filter is optimal. Such estimates can be made using an epoch (sliding window) of length N samples as:

$$y_3[n] = \frac{1}{N}\sum_{i=0}^{N-1} y_2[n-i], \tag{4}$$

where $y_3[n]$ denotes the output of the low-pass moving average filter stage. This averaging process is repeated at a series of regularly spaced sample indexes, i.e., the filter is shifted by k samples at a time ($k \leq N$). If $k = 1$, a sEMG amplitude estimate (over N samples) is produced every sample (overlap of $N-1$ samples). If $k = N$, a sEMG amplitude estimate is produced every N samples (contiguous epochs). If $k = N/2$, an sEMG amplitude estimate is produced every $N/2$ samples (50% window overlap), etc. See Fig. 5 for an example showing two different sEMG amplitude estimates using different shift (k) values. The window length N must be short enough to track the true signal changes and long enough to reduce the random fluctuations.

Alternatively, sEMG amplitude estimates can be made using a conventional low-pass filter (instead of an averaging window), including critically-damped filters (Robertson and Dowling, 2003). In this case, sEMG amplitude estimates are produced every sample and, if desired, can have their sampling rate reduced after the low-pass filter (see section 4.6.1).

4.3.2. Considerations for selection of the low-pass filter cutoff frequency

For ARV and RMS estimators (using moving average filters), selection of the epoch length, N samples, is equivalent to selecting the low-pass filter cutoff frequency, as[6]:

$$f_c = \frac{0.4429 \bullet f_s}{\sqrt{N^2-1}}, \text{ or}$$
$$N = \sqrt{\frac{0.1962 \bullet f_s^2}{f_c^2} + 1}, \tag{5}$$

where f_c is the cutoff frequency and f_s is the sampling rate (Hz). For general contractions (e.g., force-varying), the best filter type and cutoff frequency are not known theoretically. It is well known that sEMG amplitude estimate error is a trade-off of two opposing constraints. Random fluctuations (variance errors) are reduced by *decreasing* the cutoff frequency (*increasing N* in ARV and RMS estimators). However, true sudden changes in the sEMG amplitude are best tracked by *increasing* the cutoff frequency (*decreasing N* in ARV and RMS estimators).

These opposing constraints require that the cutoff frequency be selected based on the sEMG amplitude profile being studied. If fast movements are being studied, a relatively high value of f_c should be used; if slow movements are being studied, a low value should be used. Cutoff values below 1 Hz and ranging up to 10–15 Hz are typical. It is common practice to plot sEMG amplitude estimates with various low-pass filter cutoffs and overlay them over the raw sEMG data of a few exemplar contractions (Fig. 6). Visually, one can judge if the cutoff frequency is too high (allowing too much unfiltered variance) vs. too low (unable to follow true changes in sEMG amplitude).

When sEMG amplitude is time-varying, the use of averaging filters (i.e., ARV or RMS) is likely an extrapolation of their optimality when the contraction is constant-pose, constant-force and non-fatiguing; the selection is not based on a derived optimization result. A few researchers (Clancy, 1999; D'Alessio, 1984; Park and Meek, 1995; Ranaldi et al., 2018) have time-adapted the length (N) of an averaging filter. Such

[4] In order to properly estimate standard deviation, the absolute value detector *should* include a gain factor (i.e., $y_{2,abs}[n] = Gain \bullet |y_1[n]|$, whose value depends on the sEMG signal model (Clancy and Hogan, 1999). But, for the vast majority of applications, the scaling of sEMG amplitude is not important or is rendered moot by subsequent processing/normalization. Hence, this gain factor is commonly omitted.

[5] Of course, the *estimated* sEMG amplitude will show small epoch-to-epoch fluctuations due to the random nature of sEMG.

[6] The first formula is accurate to within 1% for $N \geq 4$, per https://dsp.stackexchange.com/questions/9966/what-is-the-cut-off-frequency-of-a-moving-average-filter (accessed 30 September 2022). We used MATLAB to validate this accuracy for $4 \leq N \leq 10^6$. For $N > 10$, the first formula is well approximated by: $f_c = 0.4429 \bullet f_s/N$.

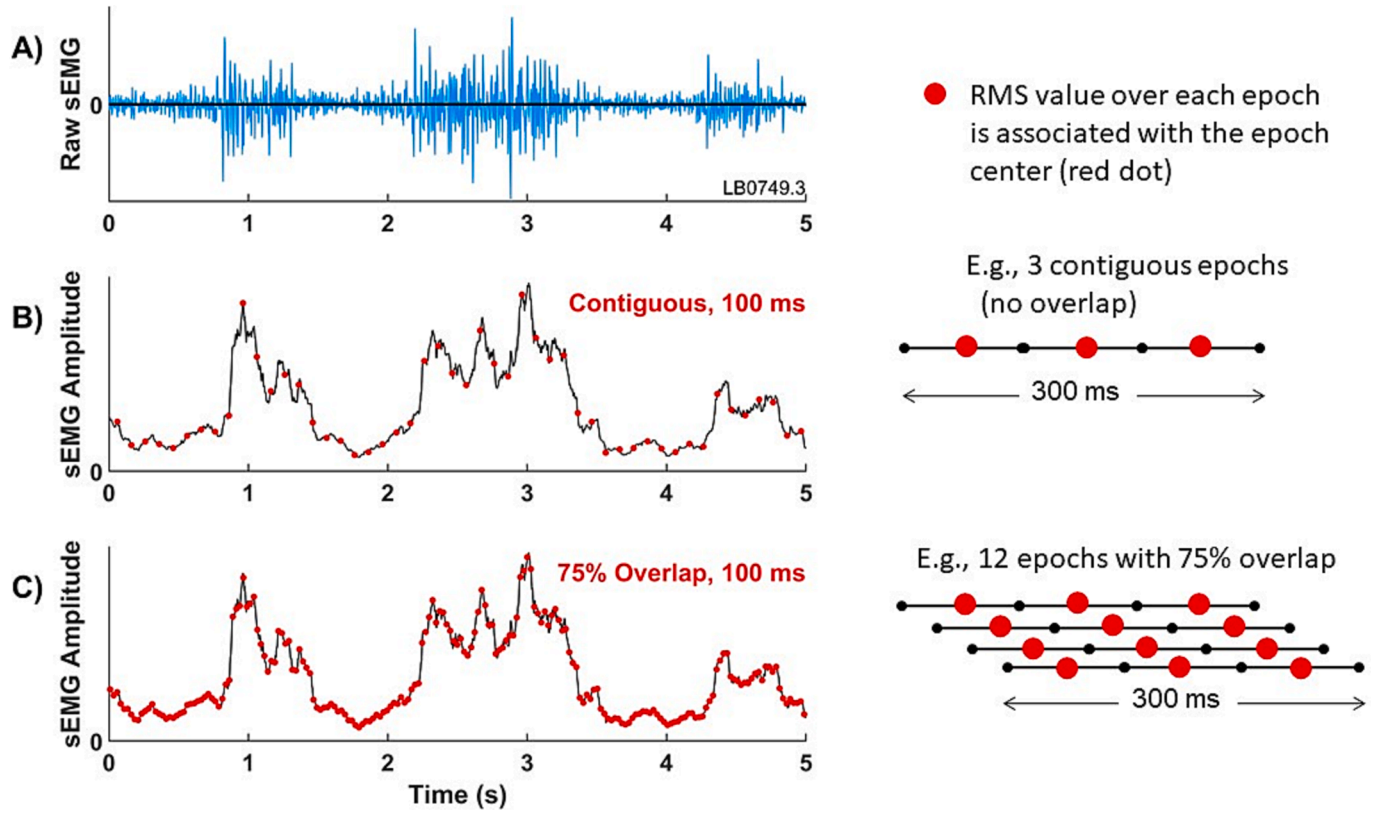


Fig. 5. Effect of epoch (sliding window) overlap. Plot A shows 5 s of a raw sEMG signal, sampled at 4096 Hz. Solid black plots in B) and C) show the sEMG amplitude estimate computed at each sample (i.e., $k = 1$), using 100 ms ARV processor. Red dotted plot in B) shows sEMG amplitude estimate using ARV processor with contiguous epochs of 100 ms duration (reducing sampling rate to 10 Hz). There are noticeable differences between the dotted red and solid black plots. Dotted red plot in C) shows similar epoch-based (a.k.a. sliding window) sEMG amplitude using 75% overlap (i.e., sliding by 25 ms each epoch; reducing sampling rate to 40 Hz) and 100 ms duration epochs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

filters are more complex to design and implement, but automatically select the optimal window duration. These filters tend to perform as well as the single best fixed window duration (and occasionally better). Conventional low-pass filter types with different cutoff frequencies have been studied (De Luca et al., 2010; Kreifeldt, 1971; Miyano et al., 1980; Taha et al., 2015; van Boxtel, 2010; Zschorlilch, 1989).

4.3.3. Low-pass filter selection in sEMG-force applications

Much caution should be taken when relating sEMG to force or torque, since the recorded sEMG is generated by only a portion of the muscle of interest, and the synergistic muscles acting about a joint and contributing to the measured torque are often neglected. Additionally, many different parameters have been extracted from sEMG and related to force individually or as an ensemble (Dai et al., 2017; Kallenberg and Hermens, 2008; Nielsen et al., 2011). Nonetheless, many biomechanics studies which relate sEMG amplitude to force use a Butterworth filter with cutoff frequency near the maximum movement frequency, i.e., $\leq$ 1–2 Hz (Brereton and McGill, 1998; Koirala et al., 2015; Manal and Rose, 2007; Olney and Winter, 1985; Potvin and Brown, 2004). This filter serves as both the low-pass filter in the sEMG amplitude processor and as the (low-pass) dynamics relating sEMG amplitude to force (Brereton and McGill, 1998; Manal and Rose, 2007; Olney and Winter, 1985; Potvin and Brown, 2004). One drawback of a Butterworth filter of order > 1 is that its output can overshoot the final value (even crossing below a zero value, which makes no physiologic sense). Thus, some researchers use a critically-damped conventional filter, which prohibits overshoot (Robertson and Dowling, 2003). Filter orders of two or higher are common. An exception is regression-based sEMG-force modeling (and other supervised modeling methods). These models use calibration trials to optimally fit the cutoff frequency and shape of the low-pass filter used to relate sEMG amplitude to force (sEMGσ-force) (Dai et al., 2019, 2017; Hajian et al., 2021). Accordingly, the preceding low-pass filter within the sEMG amplitude estimator has its cutoff frequency set to a value that is clearly *above* the dynamic movement frequencies (typically 10–20 Hz).

4.4. Relinearizer

If a square-law detector was used, then you must take the square root of the low-pass filtered signal (see Fig. 3E), to return the signal to its proper physical units (volts). This stage is omitted if the absolute value detector was used. If a moving average is used as the low-pass filter, many researchers combine the detection, low-pass filter and relinearizer stages into one filter. If an absolute value detector is used, this filter produces the ARV, a.k.a. MAV:

$$y_{4,ARV}[n] = \frac{1}{N}\sum_{i=0}^{N-1}|y_1[n-i]\,|, \tag{6}$$

where $y_4[n]$ denotes the output of the relinearizer. If a square-law detector is used, a RMS filter results:

$$y_{4,RMS}[n] = \sqrt{\frac{1}{N}\sum_{i=0}^{N-1}y_1^2[n-i]}. \tag{7}$$

4.5. Noise correction

The pre-processing stage attenuates noise outside of the sEMG bandwidth and narrow-band interference, but not broader band "background" noise that overlaps in frequency with sEMG. The sample

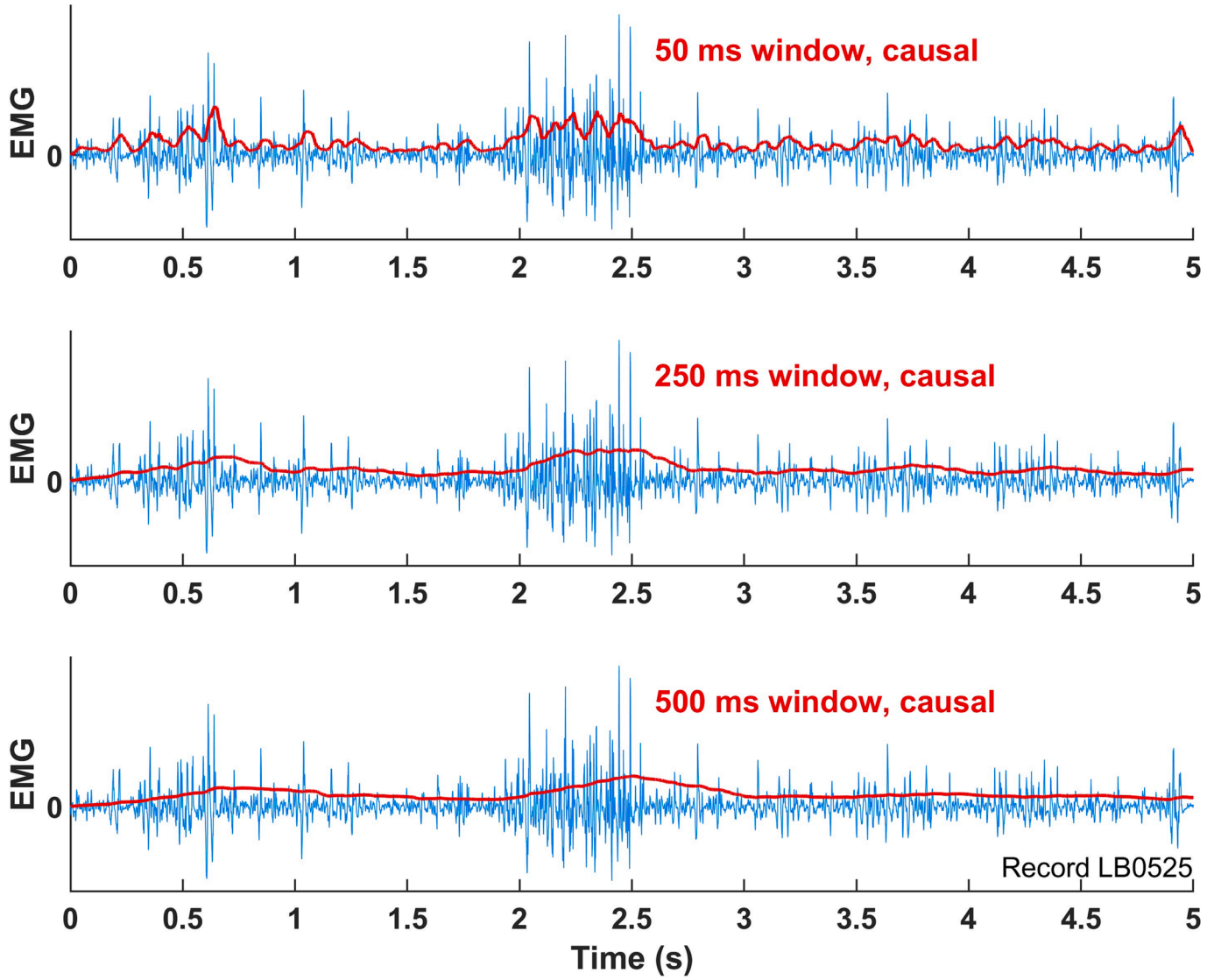


Fig. 6. All three plots show the same raw sEMG signal in blue. In each plot, a distinct sEMG amplitude estimate is shown in red, using a causal smoothing window. Top: 50 ms duration window. Middle: 250 ms duration window. Bottom: 500 ms duration window. Of the three windows tested, the 250 ms window seems best for this sEMG signal. Note the delay introduced by longer causal windows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

variance of the broader band noise, q^2, can be computed from a few seconds of rest sEMG (i.e., no contraction = 0% MVIC), *after* applying all pre-processing filters, using the sample variance formula. This noise is subtracted from the relinearized sEMG amplitude estimate at the *power level* as (Wang et al., 2019):

$$EMG\sigma[n] = \sqrt{max\left(0, y_4^2[n] - g^2q^2\right)}, \tag{8}$$

where $g = 1$ is a noise modifier (see below) and the maximum value function prevents a negative value under the square root. A complete sEMG amplitude estimate, $EMG\sigma[n]$, results (Fig. 3F).

In real-time control applications, such as using a myoelectric-controlled prosthesis, orthosis or exoskeleton, this simple correction is problematic. The user's muscles remain at rest most of the time (Weir, 2004). This noise correction would cause the device to be actuated a large fraction of the time during rest (Wang et al., 2019), which is opposed to the user's intent and could lead to unintended/dangerous device operation. Accordingly, the noise modifier is often inflated (Williams III, 1990). With $g = 1.2$, less than 5% sEMG amplitude values at rest caused inadvertent actuation in a theoretic model using a 100 ms RMS moving average. Note that excessive noise inflation can require high levels of contraction to actuate the device, leading to rapid fatigue.

4.6. Algorithm variants and comments

4.6.1. The final sampling rate of sEMG amplitude estimates

The raw sEMG signal is sampled at a rate that is much faster than that used for related slower physiological signals (e.g., joint torque or limb angle) or controlled devices (e.g., prostheses, orthoses and exoskeletons that operate at 100–200 Hz). It can be convenient to decrease the sampling rate of the sEMG amplitude estimate to match the slower rate of these other signals. When ARV or RMS smoothing is used, we have already described (section 4.3.1) how a sliding window is used to produce sEMG amplitude estimates at a lower rate, doing so in a computationally efficient manner. Each reduced-rate sEMG amplitude estimate sample represents information over an epoch. So, the new sampling rate equals the number of epochs/second.

If a conventional low-pass smoothing filter is used, sampling rate is not changed and may be later reduced through downsampling (that is, by retaining one of every k values from the original sequence and dis-

carding the others). The new sequence still represents the signal over the same time range, but at a new sampling rate equal to the original sampling rate divided by *k*. When reducing sample rate in this manner, the new sampling rate *must* be *more than twice* the cutoff frequency of the low-pass filter used after the detector.

4.6.2. Zero-phase filtering

To achieve non-causal zero-phase filtering (i.e., no delay), *each* of the filters in the entire processing chain should be implemented using a zero-phase technique. For the ARV and RMS methods of combined detection, low-pass filtering and relinearization, zero-phase filtering can be achieved by centering each epoch on the time of interest, averaging half of the past values and half of the future values. Thus, equations (6) and (7) are re-written for *N* odd-valued as:

$$y_{4,ARV,0}[n] = \frac{1}{N}\sum_{i=\frac{1-N}{2}}^{\frac{N-1}{2}} |y_1[n-i]|,$$
$$y_{4,RMS,0}[n] = \sqrt{\frac{1}{N}\sum_{i=\frac{1-N}{2}}^{\frac{N-1}{2}} y_1^2[n-i]}. \quad (9)$$

5. Surface EMG amplitude estimation of HDsEMG (high density surface EMG)

HDsEMG uses a two-dimensional (2D) grid of small electrodes (diameter $\leq$ 5 mm) to simultaneously record multiple monopolar sEMGs versus time. The electrodes of a grid should be spaced close enough to satisfy Nyquist's Criterion in *space*, hence a center-to-center interelectrode distance $<$ 5 mm should be used, corresponding to a sample frequency in space of 200 samples/meter (Merletti and Muceli, 2019). In practice, an interelectrode distance of 8–10 mm is acceptable and results in limited spatial "aliasing." Finer image resolution via 2D interpolation is permitted only if Nyquist's Criterion in space is satisfied by the original electrode spacing, so that additional electrode channels can be interpolated between existing channels. Surface EMG maps at successive time samples evolve in time, like the frames of a movie, at the original sampling rate *in time* (commonly $\geq$ 2000 Hz), which is quite high for the human eye to follow. Thus, the images must be shown at a much lower frame rate (slowed-down video).

If desired, bipolar EMG can be computed in software by differencing signals from adjacent electrodes, usually along the estimated direction of action potential propagation as shown in https://www.robertomerletti.it/en/emg/material/videos/f1. In either case (monopolar or bipolar) sEMG amplitude estimates can be computed vs. time for each epoch and each sEMG channel (image pixel). Fig. 7 shows two 8x4 grids from which 64 monopolar and 60 longitudinal single differential signals are obtained. One single differential sEMG amplitude (RMS) map is computed for one epoch. Since the epoch is 50 ms long with no overlapping, the video frame rate is 20 frames/s.

Some grid locations have larger sEMG amplitudes than others. Because of the large amount of information involved, proper interpretation of these maps requires specific training and expertise.

Fig. 7 and Fig. 8 show that the distribution of HDsEMG amplitudes can be very heterogeneous (Vieira and Botter, 2021). A grand average of the sEMG amplitudes in a map is not of great help to monitor spatial changes in the activation of a muscle because it averages values detected in different places, including the innervation and tendon zones, and even from possibly different muscles or inactive areas. Thus, it is necessary to identify regions of activity/interest from the interpolated map and limit interpretation of sEMG amplitude to a single muscle or to a portion of it. This process is called "map segmentation" and many available algorithms do it automatically (Afsharipour et al., 2016; Caselles et al., 1997; Gallina and Botter, 2013; Vieira et al., 2010; Vincent and Soille, 1991; Watanabe et al., 2013, 2012). The simplest one only uses sEMG amplitude values (pixels) above a threshold (e.g., 70% or 80% of the maximal value of the map) to estimate the region(s) of activity.

6. Estimation of sEMG features during electrically elicited contractions

Electrical stimulation of a motor nerve or muscle motor point (not to be confused with the innervation zone) by means of trains of rectangular (monophasic or biphasic) current pulses produces a train of sEMG responses called M−waves or compound motor action potentials (CMAPs), quite stable on a short-time basis since many random factors are removed. All recruited MUs discharge synchronously at the same frequency as the pulses and their number is controlled by the current level and/or duration of the stimulating pulses (Botter and Merletti, 2016; Mandrile et al., 2003; Merletti et al., 1992; Rodriguez-Falces et al., 2013). This muscle activation paradigm is very different from the way MUs are activated during voluntary contractions and is useful for investigating many nerve and muscle properties/conditions [e.g., reflexes, fatigability, spatial distribution of MUs, MU number (MUNE), nerve and muscle fiber conduction velocity, crosstalk, cramps, and response to functional electrical stimulation].

In this tutorial, only a few CMAP features are briefly addressed. CMAPs differ greatly from voluntary sEMG because the signal is almost deterministic with only small and slow random or trend changes, and because a large stimulation artifact may affect the sEMG and must be removed (Mandrile et al., 2003). Features of the CMAP that are

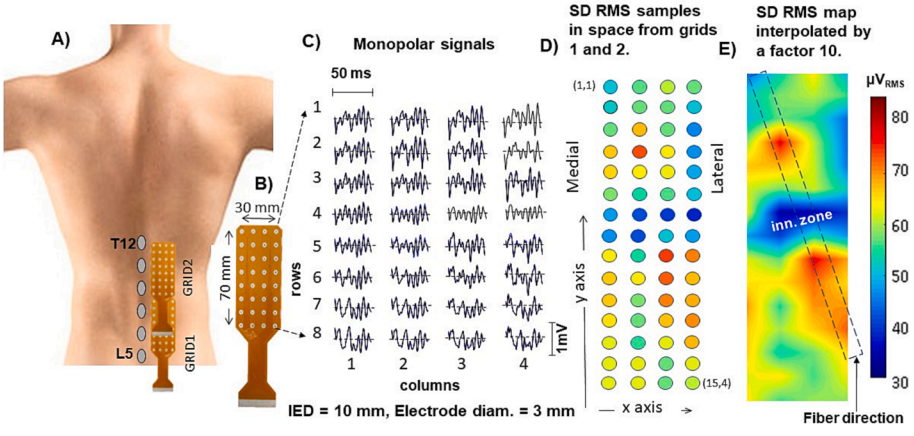


Fig. 7. HDsEMG recording from the right lumbar erector spinae with two 8x4 electrode grids with interelectrode distance (IED) of 10 mm. A) grid locations, B) grid size, C) 32 monopolar signals from grid 2 during one 50 ms epoch, D) color coded single differential (SD) sEMG amplitude (RMS) values during one epoch (color scale to the right) from the two grids, E) one video frame (interpolated by a factor 10) showing sEMG amplitude (RMS) distribution in space under the two grids. Note the possibility to identify the innervation zone (blue area) and the fiber direction of the muscle. This figure is best interpreted when viewed in color. (Modified, with permission, from https://www.robertomerletti.it/en/emg/material/teaching/module7.) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

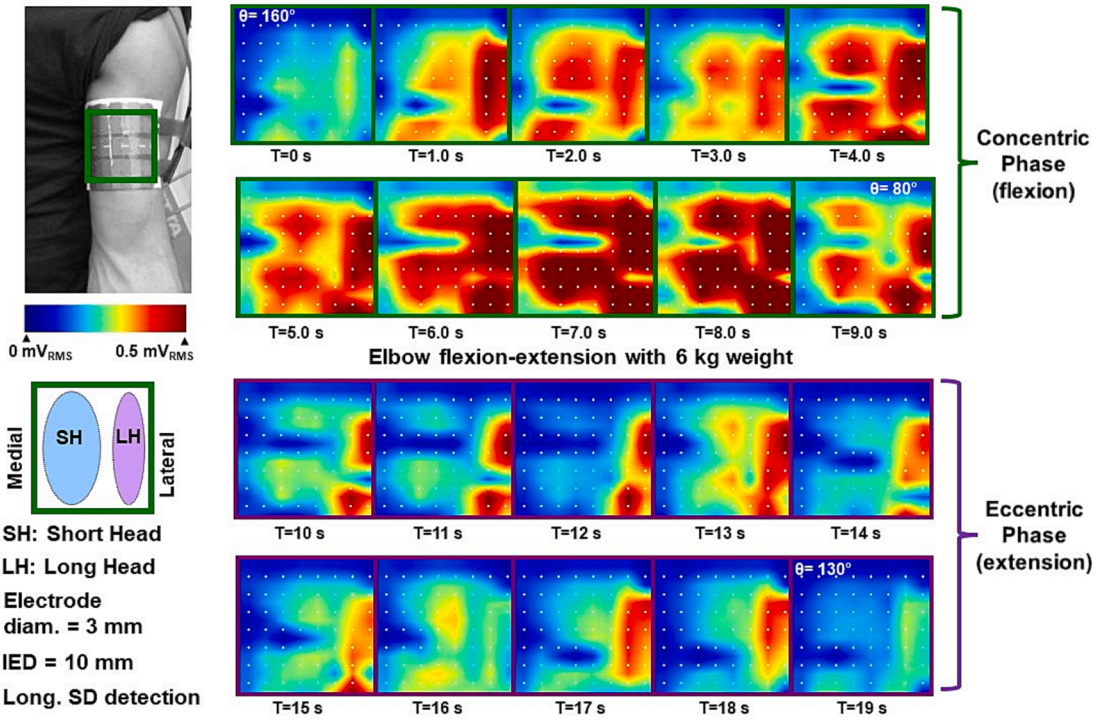


Fig. 8. Example of interpolated longitudinal single differential (SD) sEMG amplitude maps detected from the biceps brachii. The subject held a 6 kg weight in his left hand and slowly flexed the elbow from 160° to 80° in 10 s (concentric phase) and then extended it back to 130° (eccentric phase) in 10 s. Surface EMG amplitude maps were computed every 1 s and interpolated (by a factor of 10). White dots indicate the electrodes. Note the different role and behavior of the long and short head of the muscle during the two phases. Note that the upper row of the electrode grid is on the muscle–tendon junction. Note the shifts of the innervation zones of the long and short heads as the forearm moves. IED = interelectrode distance. This figure is best interpreted when viewed in color, as is available in the web version of this article. (Modified, with permission, from https://www.robertomerletti.it/en/emg/material/teaching/module7; see video at https://www.robertomerletti.it/en/emg/material/videos/f10).

commonly measured over time are: a) peak-to-peak value, b) positive and negative peak values, c) areas under the positive or the negative peaks, d) RMS of the full wave or of a wave train, and e) duration of the positive and negative phases. Other features are measured in the frequency domain. Since CMAP duration may reach 30–35 ms, repetitive stimulation should be below 30 pulses/s (pps) to avoid overlapped responses. The spatial distribution of CMAPs may be investigated with HDsEMG arrays. A particularly interesting application is the differential CMAP, obtained by subtracting two CMAPs (or CMAP images) obtained in response to stimulation pulses of slightly different current level. This technique provides the contribution of the few (possibly single) MUs recruited by the second pulse and not by the first. Care must be taken concerning the application and documentation of both the stimulation and detection systems, which must be electrically insulated from each other.

7. Recommendations for best practices

Surface EMG amplitude is one of the many pieces of information (including spectral information, muscle fiber conduction velocity, MUAP features, etc.) used to extract muscle properties and is affected by both central and (mostly) peripheral factors including anatomy, physiology, and details of the detection system (Farina et al., 2002; Vigotsky et al., 2018). Hence, understanding muscle properties is usually impossible on the basis of sEMG amplitude alone.

A compromise in electrode placement may be necessary between selectivity (detection volume) and crosstalk from other muscles. Proper measurement and interpretation of sEMG amplitude changes and trends require normalization when comparing across muscles or subjects (Besomi et al., 2020; Burden, 2010). All these considerations require clinical time, familiarity with instrumentation, training, competence and care in performing measurements. For these reasons, sEMG is viewed with perplexity and diffidence by many clinical operators (Campanini et al., 2020, p. 202; Merletti et al., 2021; Vigotsky et al., 2018). Best practices must be followed to teach the methodology and allow comparison of results.

It is highly recommended to retain and visually analyze the raw sEMG because it may contain artifacts, interferences or spurious components that will be obscured once sEMG amplitude is estimated (including interference from room lighting, equipment power supplies, etc.). These sources may be intermittent. Most can be readily visually identified and the raw sEMG, if available, could be reprocessed differently to obtain a better quality signal. For example, a task requiring limited, slow movement might have the raw sEMG filtered with a low-order high-pass filter. However, if the muscle is occasionally perturbed mechanically (e.g., "bumped") or involved in a rapid trajectory correction, movement artifacts may be produced that are not adequately removed by the low-order high-pass filter. Visualization of the raw sEMG may distinguish such deflections from actual muscle activity. In addition to visualizing the raw sEMG, examining the power spectral density of the raw sEMG can identify narrow-band interference sources (such as power-line interference).

7.1. Reporting geometry and location of the detection system

Monopolar or bipolar detection: A first fundamental issue concerns the selection of the detection technique (monopolar, bipolar, or electrode arrays) [see (Besomi et al., 2020, 2019; Gallina et al., 2022; Merletti and Muceli, 2019)]. Justification for the detection technique and its geometry as well as for the selected normalization technique must be provided. The noise and interference magnitudes and the voltage range of raw or pre-processed signals should always be reported as referred to the

input of the front-end amplifier (e.g., in physical units of μV or mV), not to its output.

As indicated in previous tutorials, the user must be knowledgeable that sEMG amplitude is actually a 2D signal, that is an image or map, showing the non-uniform distribution of instantaneous sEMG amplitude in space above the muscle(s) of interest. This map is monopolar but detection may use a spatial filter, typically a bipolar or single differential (SD) configuration which introduces a "transfer function" that filters the signal and modifies its amplitude and spectrum (Merletti and Muceli, 2019). The geometry (size and interelectrode distance) and location (with respect to the innervation zone or anatomical references) of an electrode pair have significant impacts on the detected sEMG amplitude (Farina et al., 2002) and therefore must be reported and must be the same when measurements are repeated on the same muscle(s) of the same subject. The size and location of the reference electrode (if used) should also be indicated. Surface EMG amplitude measurements performed on different muscles or different subjects must be compared with great caution and after normalization, usually with respect to MVIC (Besomi et al., 2020; Minetto et al., 2013).

Skin treatment and the use of conductive gel (and its manufacturer) should be mentioned as well as any other detail required for reproduction of the tests under the same conditions. Use of alcohol to clean the skin may not lower the contact impedance (Piervirgili et al., 2014). Some authors report the electrode–skin impedance, but fail to indicate how it was measured. The noise and interference magnitudes and the voltage range of raw or pre-processed signals should always be reported as referred to the input of the front-end amplifier (e.g., in physical units of μV or mV), not to its output. Electrode impedance is a function of frequency. Some authors measure and report its value. In this case, the modality and frequency used for the measurement should be reported[7]. When modern high input-impedance amplifiers are used, this parameter does not significantly affect sEMG amplitude, but electrode–skin impedance unbalance enhances power-line interference. Some manufacturers report the input *resistance* of their amplifiers (i.e., impedance at DC or 0 Hz) which is much greater than the impedance at the power-line frequency (5–10 GΩ instead of 0.1–0.5 GΩ). This practice is misleading because the input impedance of the amplifier must be high at the power-line frequency (50 Hz or 60 Hz), not at DC (0 Hz), to reduce interference affecting sEMG amplitude estimation (Merletti and Cerone, 2020).

HDsEMG detection: The considerations above apply to HDsEMG recordings as well. When electrode arrays are used it is important to verify, by means of preliminary test contractions, that all electrodes are in good contact with the skin and the signals are not affected by interference. Verification can be achieved by looking at the sEMG amplitude map; spots very different from their neighbors are suspicious. The number of rows and columns of the grid should be reported as well as the orientation of the array with respect to the estimated direction of the muscle fibers. A color bar should indicate the correspondence between colors and raw sEMG or sEMG amplitude voltages referred to the input of the amplifier, as shown in Fig. 7 and Fig. 8.

7.2. Reporting signal conditioning features, background noise and front-end amplifier saturation

Surface EMG amplitude is affected by signal conditioning and pre-processing methods such as low- and high-pass filters, and techniques used for removal of power-line interference. Although the technical specifications of the front-end amplifier(s) may be available from the manufacturer's website, it is good practice to report the main ones (bandwidth, input impedance and CMRR at the power-line frequency, gain, input referred noise) (Merletti and Cerone, 2020).

It is good practice to record sEMG under relaxed/rest conditions, to measure background noise and verify the absence of interference or identify the type of interference present, and to measure MVICs to confirm the absence of amplifier saturation. Verification of absence (or removal) of interfering physiological signals (e.g., ECG) should be reported when dealing with trunk or back muscles. Methods to reduce it should be described or cited.

7.3. Measurement of sEMG amplitude during isometric constant force contractions

During isometric constant force (isotonic) contractions of low or intermediate level, sEMG amplitude changes slowly, because of progressive fatigue or changes of the load sharing, and is almost unchanging over time intervals of 0.5–2 s. If sEMG amplitude is estimated by the ARV or RMS methods (Fig. 2), the respective estimation method should be indicated, as well as epoch length and overlap of the moving window. Else, the detector type must be specified (absolute or square) and the ensuing low-pass smoothing filter fully described (e.g., the filter type, order, and cut off frequency). In either case, causal or non-causal (zero-phase) filter implementation should be noted (otherwise, causal is assumed). Window (epoch) lengths of 0.5–2 s or cutoff frequencies of the low-pass filter of 0.25–1 Hz are common for this case. The result is a slowly varying and slightly fluctuating estimate versus time. For high or maximal contraction levels, shorter windows or higher cutoff frequencies better track the trend and fluctuations of the estimated sEMG amplitude (windows of 0.25–1 s and cutoff frequencies of 0.5–2 Hz are common). When a moving average window is used, the value of the amplitude should be attributed to the center point of the window (Fig. 5).

7.4. Reporting sEMG amplitude during isometric force-varying or dynamic contractions

During isometric force-varying contractions (Fig. 1 and Fig. 3) all the considerations provided in sub-section 7.3 apply but the window length, or the cutoff frequency of the low-pass filter, must allow for tracking the fast amplitude modulation of the signal. For this reason, window lengths of 50–300 ms, or low-pass filter cutoff frequencies of 2–10 Hz, are used to avoid excessive smoothing of the envelope. Consider that the rise and fall time of the sEMG amplitude is influenced largely by the filter used and not only by the sEMG burst properties, as indicated in Fig. 6 and Fig. 9. Adaptive moving average filters that automatically modify the window length of ARV or RMS processors have been investigated, as mentioned in subsection 4.3.2. If utilized, full methods and/or appropriate references should be provided.

The same considerations apply to dynamic contractions with one important additional consideration. Since the muscle shortens or lengthens under the skin, the relative position of the electrodes with respect to the innervation zone changes causing fluctuations of the bipolar sEMG amplitude estimates [see Fig. 8 herein and Fig. 17 of (Merletti and Muceli, 2019)]. This may cause misinterpretation of the relationship between sEMG amplitude and joint angle. It is good practice, whenever possible, to be aware of the location and shift of the innervation zone due to the muscle's movement (Barbero et al., 2012). This problem is somewhat reduced if the double differential detection method is used, because this modality does not show a minimal amplitude when the electrodes are above a narrow innervation zone.

Care must be taken in the study of explosive or ballistic contractions when sEMG amplitude is changing rapidly and the bursts are of short duration with a fast rise time. Testing (and reporting) different smoothing parameters may be appropriate in this case. Care must be taken because of the transients introduced by filtering (Fig. 6 and Fig. 9) and the modalities of sEMG amplitude estimation should be justified. The extrapolation of the rate of increase of sEMG amplitude to the rate of development of muscle force or joint torque also requires caution because of the different transients introduced by low-pass filtering

[7] See: https://www.robertomerletti.it/assets/pdfs/technical_notes/technical_note_1_concept_of_impedance.pdf.

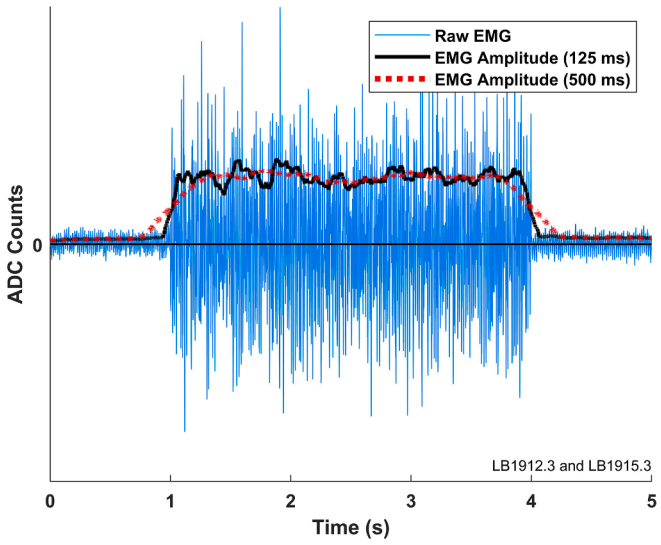


Fig. 9. Step change in raw sEMG between 0% MVIC (0–1 s), 50% (constant-posture, constant-force) MVIC contraction (1–4 s) and 0% MVIC (4–5 s) shown in blue. Non-causal (zero-phase) ARV sEMG amplitude estimate shown in solid black (125 ms window) and dotted red (500 ms window), both producing an estimate at each sample (maximum overlap). The 500 ms window (large smoothing) causes sEMG amplitude change to spread across a large time duration at each step change. Because a non-causal (bidirectional) estimate was used, each sEMG amplitude estimate begins changing before the step occurs and ends after the step terminates. In general, a 500 ms window is too large to track this rapid change in sEMG amplitude. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(Fig. 9) and by the excitation–contraction process within muscle(s).

7.5. Integrated sEMG

Some old analog equipment integrated the rectified sEMG until a threshold was reached. The integrator was then reset and a pulse generated. By counting the pulses over a given time interval the area under the rectified sEMG signal was obtained. This old technique is virtually abandoned and is not recommended.

7.6. Reporting HDsEMG amplitude

The HDsEMG array of signals provides a time varying map of raw instantaneous sEMG signals (one map per time sample). The considerations provided in previous subsections for single channel sEMG apply to each pixel or channel. A sEMG amplitude map is defined for each epoch. Since only parts of the map may be of interest, proper criteria must be used and described for the segmentation of the map to obtain regions of activity. To obtain a spatial average of the sEMG amplitude for each region of activity, the single pixel values are averaged over the pixels contained in each region (spatial average). In turn, this spatial average, estimated for each epoch, becomes a new signal providing an estimate of average sEMG amplitude over the region versus time. Of course, care must be taken in case the area or position of the region of activity changes over time, indicating changes of intensity or position of the active portion of the muscle(s) and/or different load sharing among muscles. The centroid of each region has been used to provide a rough but synthetic indication of activity localization (Gallina et al., 2016; Russo et al., 2019, p. 2021; Vieira and Botter, 2021; Vieira et al., 2010).

The user should be aware that large electrode arrays may cover an area with non-uniform thickness of subcutaneous tissue. The sEMG amplitude of pixels relatively far from each other may therefore reflect both the muscle activation and the depth of the sources under the skin. Measurement of skinfold with a plicometer or, and better, by echography would provide indications concerning fat layers and pennation angle. In addition, some portions of a pennate muscle under the array may appear momentarily inactive while they are not (dos Anjos et al., 2017).

7.7. Reporting sEMG features during electrically elicited contractions

Electrically elicited contractions provide almost deterministic sEMG signals whose features can be measured more easily and reliably than in the case of voluntary signals. Considerable information can be obtained from them if the stimulation artifact is negligible or is properly removed without altering the signal. The features of the stimulator (controlled current or controlled voltage or “hybrid”) must be reported (Botter and Merletti, 2016; Mandrile et al., 2003; Merletti et al., 1992) as well as the pulse waveform, frequency and duration, and the stimulation electrode size, shape, distance and location. The same considerations described above for the sEMG detection system apply with the addition of reporting the hardware or software modalities for artifact suppression and the features of the CMAP being measured.

7.8. Checklist for preparing and reviewing reports and manuscripts dealing with sEMG amplitude estimation

Table 1 provides a checklist for preparing and reviewing reports and manuscripts dealing with sEMG amplitude estimation.

8. Concluding remarks

This tutorial focuses on some fundamental concepts required to critically understand or write scientific works involving sEMG amplitude estimation. It deals with only one of the many features of sEMG and should be complemented by tutorials on other features (e.g., spectral features and conduction velocity). Inappropriate extrapolation of sEMG amplitude information to extract anatomical/physiological properties of muscle can be very misleading and dangerous. Warnings and examples of information that *cannot* be extracted from amplitude estimates of the voluntary sEMG or require different approaches include (Vigotsky et al., 2018):

1 Changes of MU firing rate and recruitment strategies *cannot* be discriminated using sEMG amplitude only.

2. Because of the attenuation in signal magnitude due to MU depth, recruitment of larger and deeper MUs may have an effect similar to the recruitment of smaller but more superficial MUs. Therefore, in general, MU location and depth cannot be estimated using sEMG amplitude.

3 A muscle can produce force with a small or zero sEMG amplitude due to its passive properties (e.g., stretching by other muscles).

4. A large sEMG amplitude may be associated with small or zero net torque at a joint because of coactivation of antagonist muscles. Simultaneous measurement of agonist and antagonist sEMG amplitude is recommended.

5. Surface EMG may not reliably predict muscle force during dynamic tasks, even after normalization, and may be affected by the sliding of the innervation zone under the electrodes.

6. Surface EMG amplitude detected from a *single* muscle does *not* usually reflect torque at the joint because of possibly changing load sharing with other synergic and antagonist muscles.

Different degrees of care and caution must be used if sEMG amplitude measurements are carried out within- vs. between-subjects, within- vs. between-muscles, and all combinations thereof (Vigotsky et al., 2018). Studies investigating the predictive validity of sEMG amplitude are lacking (Halperin et al., 2018). The amplitude of the sEMG signal is one of the many features of the signal and is just a piece of a jigsaw puzzle. In a few cases (e.g., biofeedback) it may be sufficient but, in general, ambiguity is reduced by measuring other features as well.

Replicability of experiments is at the heart of science. Reporting

Table 1
Checklist for preparing and reviewing reports and manuscripts dealing with sEMG amplitude estimation.

Topic/item	Description and information to report	Considerations, comments and references
1. Detection modality	1.1. Wired or wireless systems 1.2. Monopolar, bipolar single differential, or other spatial filters. Arrays (HDsEMG)	See tutorial (Merletti and Muceli, 2019)
2. Electrode geometry and location	2.1. Electrode size, shape and interelectrode distance, orientation 2.2. Location with respect to innervation zone oranatomical references 2.3. Size and orientation of the array 2.4. Location of the reference electrode 2.5. Skin treatment	See tutorial (Merletti and Muceli, 2019) and consensus paper (Besomi et al., 2019). Report manufacturer of electrodes, gel, etc. Discourage placing electrode pairs over muscle belly as that location is often the innervation zone—a point of minimal sEMG amplitude for bipolar detection (Barbero et al., 2012)
3. Front-end amplifier, analog signal conditioning, and ADC conversion	3.1. Brand and model 3.2. Main technical specifications of amplifier 3.3. Analog filter order, type and bandwidth 3.4. ADC converter specifications	See tutorial (Merletti and Cerone, 2020) and manufacturer specifications: input impedance and CMRR at 50–60 Hz, RTI noise, number of bits and range of ADC
4. Experiment pre-trial checks	4.1. Reduction of power-line, artifacts and ECG interference before ADC data storage 4.2. Record noise amplitude in relaxed condition 4.3. Report visual check for signal quality	Report methods/circuitry/ algorithms used, if any. Report or plot raw sEMG showing signal quality and residual artifacts and interferences
5. Muscles and their contraction modality	5.1. Contraction type 5.2. Non-monitored agonist or antagonists	Contraction type: isometric (constant force or not), dynamic (range of motion), likelihood of geometrical effects due to muscle movement
6. Methods for sEMG amplitude estimation (envelope definition)	6.1 Digital pre-processing & signal conditioning filter specifications, including interference removal 6.2a. If ARV, RMS: epoch duration, overlap (in seconds or percent, if any) 6.2b. Else: Detector (absolute vs. square), smoothing low-pass filter specifications 6.3. Details of RDS noise correction	Report technical specifications of all filters: Type and order, cutoff frequency, note if zero-phase, and any other parameters. Report how variance of rest noise was computed for RDS noise correction
7. Amplitude of HDsEMG	7.1. Same sEMG amplitude information as item 6 7.2. Definition of ROAs (regions of activity) 7.3. Dimensionality reduction methods (averaging over ROAs, indication of centroids, others)	See consensus paper (Gallina et al., 2022)
8. Normalization method	8.1. Report/discuss type of normalization used, if any	See consensus paper (Besomi et al., 2020)
Electrically elicited contractions:		
9. M−wave or compound muscle action potential (CMAP)	9.1. Features of the stimulator 9.2. Pulse waveform, duration, frequency 9.3. Stimulation electrode shape, size and location 9.4. Detection electrode shape, size and location 9.5. Technique used for stimulation artifact reduction	Stimulator may control current, voltage or be hybrid. See items 1–4 for detection electrode requirements. See (Botter and Merletti, 2016; Mandrile et al., 2003; Merletti et al., 1992)

information sufficient to exactly reproduce a test or a procedure is a fundamental good practice.

Institutional review board and informed consent statement

The data shown in Figs. 1, 3–6 and 9 are de-identified and exempted from human studies supervision per Worcester Polytechnic Institute (WPI) Institutional Review Board letter 10–100. A full description of the original experimental procedures, including a statement that all subjects provided written informed consent, first appeared in (Clancy and Farry, 2000). The data shown in Figs. 7 and 8 were reused, with permission, from on-line teaching material (https://www.robertomerletti.it/assets/pdfs/comes_rapporto_finale.pdf) prepared within the Italian teaching project CoMES (Motor Control and sEMG: development of models and teaching tools), approved by the ethical committee of the University of Torino, Italy. All subjects provided informed consent.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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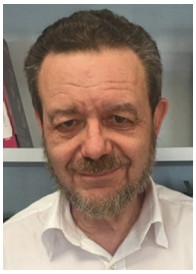
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